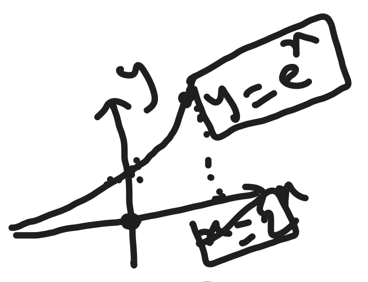

$(x - 2)$ $x_0$ 0

## Taylor and Maclaurin Series:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

**Definition**: If $f$ has derivatives of all orders at $x_0$, then we call the series

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

$$= f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f'''(x_0)}{3!}(x - x_0)^3 + \frac{f^{iv}(x_0)}{4!}(x - x_0)^4 + \cdots$$

the Taylor Series for $f$ about $x = x_0$

If the special case where $x_0 = 0$, this series becomes,

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{iv}(0)}{4!}x^4 + \cdots$$

in which case we call it the Maclaurin series for $f$.

**Ex:** Find the Taylor Series for $\frac{1}{x}$ about $\boxed{x = 1}$

**Sol$^n$:** $f(x) = \frac{1}{x}$, $x_0 = 1$

| | |
|---|---|
| $f(x) = \frac{1}{x} = x^{-1}$ | $f(1) = 1$ |
| $f'(x) = -\frac{1}{x^2} = -x^{-2}$ | $f'(1) = -1$ |
| $f''(x) = 2x^{-3}$ | $f''(1) = 2$ |
| $f'''(x) = -6x^{-4}$ | $f'''(1) = -6$ |
| $f^{iv}(x) = 24x^{-5}$ | $f^{iv}(1) = 24$ |
| $f^{v}(x) = -120x^{-6}$ | $f^{v}(1) = -120$ |

$$f(x) = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 + \frac{f^{iv}(1)}{4!}(x-1)^4 + \frac{f^{v}(1)}{5!}(x-1)^5 + \cdots$$

$$= 1 - (x-1) + (x-1)^2 - (x-1)^3 + (x-1)^4 - (x-1)^5 + \cdots$$

[Power series of $(x^2+2)$, $x_0 = -2$]

**Ex:** Find the Maclaurin Series for the function $\ln(1+2x)$.

**Sol$^n$:** $x_0 = 0$

| | |
|---|---|
| $f(x) = \ln(1+2x)$ | $f(0) = 0$ |
| $f'(x) = \frac{1}{1+2x} \cdot 2 = 2(1+2x)^{-1}$ | $f'(0) = 2$ |
| $f''(x) = -2(1+2x)^{-2} \cdot 2 = -4(1+2x)^{-2}$ | $f''(0) = -4$ |
| $f'''(x) = 8(1+2x)^{-3} \cdot 2 = 16(1+2x)^{-3}$ | $f'''(0) = 16$ |
| $f^{iv}(x) = -96(1+2x)^{-4}$ | $f^{iv}(0) = -96$ |

4
×12
−96
4×3×2

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{iv}(0)}{4!}x^4 + \cdots$$

$$= 0 + 2x - 2x^2 + \frac{8}{3}x^3 - 4x^4 + \cdots$$

## Rolle's theorem

**Statement:** Let $f$ be differentiable on $(a, b)$ and continuous on $[a, b]$. If $f(a) = f(b)$, then there is at least one number $c$ in $(a, b)$ such that $f'(c) = 0$.